
Subject-Level Calibration Heterogeneity is Stable

Across LLM Families and Scales: A Statistical Audit of MMLU

Anonymous Author(s)

Anonymous Institution

Anonymous Address

anonymous@email.com

1 Problem and Motivation

Calibration is nearly always evaluated as a single aggregate scalar—ECE, NLL, or related—that aggregates over all subjects in a benchmark, obscuring whether miscalibration is diffuse or concentrated in specific domains. We show the concentration is structured: across seven instruction-tuned LLMs from three families, the same subjects are stably worst-calibrated (college-level technical content, professional law, virology) and best-calibrated (high-school Social Sciences, applied/factual subjects), with subject-level ECE rankings correlated at Spearman $\rho \in [0.80, 0.94]$ across all 15 pairs of larger models. This stability points to task structure rather than any particular training pipeline. The distinction matters for practice: if the pattern is stable, a subject-aware audit run on one model generalizes to others, enabling portable pre-deployment diagnostics without re-establishing baselines. Existing work has noted calibration degrades on professional knowledge tasks [Xiong et al., 2024], but no prior work formally audits this structure—testing which subjects are robustly miscalibrated, how much variance they explain, and whether the pattern holds across models.

2 Approach

We apply a statistical audit framework to MMLU [Hendrycks et al., 2021] across seven instruction-tuned models: GPT-4o-mini, Llama-3-8B-Instruct, and Qwen-2.5-{0.5B, 1.5B, 3B, 7B, 14B}-Instruct. Confidence is elicited via softmax-renormalized token-level logprobs over {A, B, C, D}. The framework combines: (i) a three-level mixed-effects model decomposing calibration-gap variance across questions, subjects, and domains; (ii) isotonic reliability curves and ECE per subject; (iii) Benjamini–Hochberg FDR correction [Benjamini and Hochberg, 1995] across 57 subjects per model; and (iv) Spearman rank correlations of subject-level ECE across all 21 model pairs. Question-level covariates (word count, entropy, negation, option length) are included as fixed effects to characterize which item features predict miscalibration.

3 Results

Structured heterogeneity. Between-subject variance accounts for 1.7–7.9% of total calibration-gap variance (ICC) across models—small in absolute terms but concentrated in stable subject clusters. Gap ranges span 3.8–13.0 $\times$ from the best- to worst-calibrated subject across the six larger models. At FDR $\alpha = 0.05$, five of six larger models reject all 57 subjects; the heterogeneity is statistically real and robust to bin count and proper-scoring alternatives.

Stable patterns across models. Figure 1 shows the 7×7 Spearman rank correlation matrix of subject-level ECE. Among the six models above 1B parameters, all 15 pairwise correlations fall in $[0.80, 0.94]$. Cross-family correlations (GPT-4o-mini vs. Qwen-7B: 0.92; GPT-4o-mini vs. Qwen-14B: 0.91) are comparable to the strongest within-family pairs, indicating the pattern is task-driven rather than family-specific. Qwen-0.5B is the conspicuous exception ($\rho \in [0.13, 0.20]$ with the others), driven by extreme positional bias rather than genuine knowledge gaps.

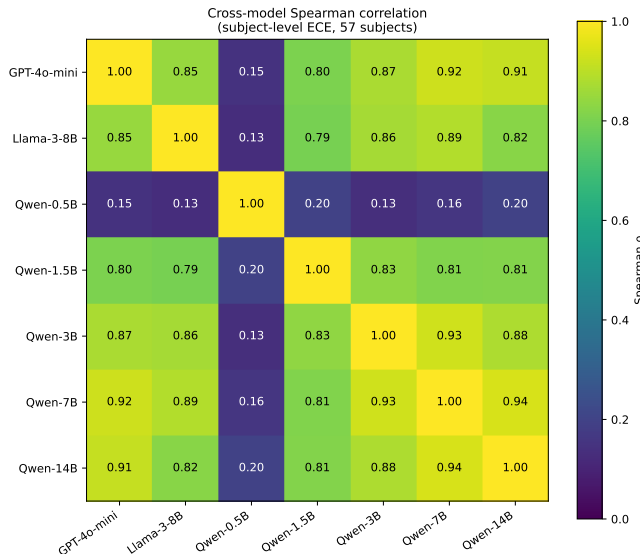


Figure 1: Spearman rank correlation of subject-level ECE across seven models. Among the six larger models, all 15 pairs fall in $[0.80, 0.94]$. Qwen-0.5B’s low correlations reflect positional bias, not calibration structure.

Worst-calibrated: virology and professional_law land in the top-10 for every larger model; abstract_algebra, college_mathematics, and machine_learning each appear for four of six. **Best-calibrated:** marketing and high_school_psychology are consistently at the other extreme. The clearest organizing axes are specialization level (high-school vs. college content) and domain category (STEM vs. Social Sciences). **Scaling:** within the Qwen family, the entropy coefficient on the calibration gap grows monotonically with scale ($-0.004 \rightarrow +0.006 \rightarrow +0.029 \rightarrow +0.040$, significant at 7B and 14B), with confidence staying near saturation as accuracy drops on harder items—a pattern also documented at the aggregate level for RLHF-tuned models [Kadavath et al., 2022, Nakkiran et al., 2025].

4 Key Contribution

We provide the first formal statistical audit of subject-level calibration heterogeneity in LLMs: variance decomposition, FDR-corrected rankings, and cross-model stability testing across seven models from three families. The finding that subject-level ECE rankings are highly stable across families and scales has two practical consequences. First, the audit is portable: a deployer can apply it to a new model and read off subject-level overconfidence directly, using the recurring offenders (professional_law, virology, college-level technical content) as a pre-deployment checklist. Second, the stability provides the empirical precondition for subject-aware recalibration: if the same subjects are consistently worst-calibrated, there is structure worth correcting at the subject level rather than globally.

References

Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society, Series B*,

- 62 57(1):289–300, 1995.
- 63 Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and
 64 Jacob Steinhardt. Measuring massive multitask language understanding. In *Proceedings*
 65 *of the International Conference on Learning Representations*, 2021.
- 66 Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez,
 67 Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, et al. Language
 68 models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.
- 69 Preetum Nakkiran, Arwen Bradley, Adam Goliński, Eugene Ndiaye, Michael Kirchhof, and
 70 Sinead Williamson. Trained on tokens, calibrated on concepts: The emergence of semantic
 71 calibration in LLMs. *arXiv preprint arXiv:2511.04869*, 2025.
- 72 Miao Xiong, Zhiyuan Hu, Xinyang Lu, Yifei Li, Jie Fu, Junxian He, and Bryan Hooi.
 73 Can LLMs express their uncertainty? An empirical evaluation of confidence elicitation in
 74 LLMs. In *Proceedings of the Twelfth International Conference on Learning Representa-*
 75 *tions*, 2024.